Quantitative mixed-state entanglement for anomalous Abelian boundaries

We present candidate exact entanglement results for specified lattice representatives of anomalous symmetries. For a CZX-type action on any Eulerian graph, a parity of the edges crossing a tripartition determines the sharp weighted convex-roof minimum in each charge sector. On finite-range chains this parity is the $\mathbb{Z}_2$ endpoint anomaly. For the type-I $\mathbb{Z}_N$ domain-wall action with nonzero class p, every charge-sector projector is negative under partial transpose across every connected bipartition when $N \geq 3$; for $N = 3$ its trace norm is exactly $(5+8\cos(\pi/9))/9$. These projectors have maximally mixed proper subsystems. The latter result rules out a complete bipartite-product eigenbasis for this representative, contrary to a broad expectation in Lessa, Cheng, and Wang, without contradicting their theorems. We also give an explicit, size-independent tripartite lower bound for the $\mathbb{Z}_N$ actions. These are representative-specific results; specialist verification and priority review remain open.

The obstruction to localizing an anomalous symmetry provides constraints on the entanglement of its states. For one-dimensional bosonic boundaries, Lessa, Cheng, and Wang [1] proved that strongly symmetric states are tripartite nonseparable and established a separate finite-depth preparation obstruction. They conjectured a sharp, size-independent value for a convex-roof measure of the CZX action [2] and gave candidate attaining bases. They also expected their construction of a complete bipartite-product eigenbasis to extend to finite Abelian groups. We give quantitative calculations for two concrete families. The first extends the CZX calculation to Eulerian graphs; the second supplies a counterexample to the broad reading of that basis expectation. It leaves open the weaker possibility that a different representative admits such a basis.

For a pure state of three parties let

$$\mathcal{F}_w(\psi) = \sum_{R=A,B,C} w_R S(\rho_R), \qquad w_R \geq 0, \tag{1}$$

and extend it to mixed states by the pure-state convex roof. Here $\rho_R = \mathrm{Tr}_{R^c} |\psi\rangle\langle\psi|$ and logarithms are natural. The measure $E_3 = \mathcal{F}_{(1/2,1/2,1/2)}$ is that used in Ref. [1]; see also Ref. [3]. Strong symmetry means support in a single eigenspace, not merely commutation with the symmetry.

Eulerian graph result.— Let $G = (V, E)$ be a finite undirected loopless Eulerian graph with qubits on its vertices. Parallel edges may be counted with multiplicity. Define the Hermitian involution

$$U_G = i^{|E| \bmod 2} \Big(\prod_{v \in V} X_v \Big) \Big(\prod_{uv \in E} \mathrm{CZ}_{uv} \Big). \tag{2}$$

For three disjoint nonempty parts A, B, C covering V, let e_{RS} count edges between R, S modulo two. Every Eulerian cut has even degree; therefore $e_{AB} = e_{BC} = e_{CA} =: e$. Put

$$p_* = (2+\sqrt{2})/4, \qquad W = \sum_R w_R,$$
$$B_w = \min\{(W - \max_R w_R) \log 2,\, W h(p_*)\}, \tag{3}$$

where $h(p) = -p \log p - (1-p)\log(1-p)$. We obtain

$$\mathcal{F}_w(\rho) \geq e B_w, \qquad \mathcal{F}_w\left(\frac{\mathbb{I} + \lambda U_G}{2^{|V|}} \right) = e B_w, \tag{4}$$

for $U_G \rho = \lambda \rho$, $\lambda = \pm 1$. If $e = 0$, the sector projectors have complete product bases across the specified partition. If $e = 1$, both branches in Eq. (3) have explicit attaining bases. For $e = 1$, each sector projector has $E_3 = \frac{3}{2} h(p_*) = 0.901314055\ldots \log 2$; for $e = 0$ this value is zero. For a nearest-neighbor ring cut into three connected arcs the $e = 1$ value is the value conjectured in Ref. [1], which already provided the balanced candidate basis.

Proof of Eq. (4).— The unphased action sends a bit string b to its complement with phase $(-1)^{q(b)}$, where $q(b) = \sum_{uv \in E} b_u b_v$. Its square is $(-1)^{|E|} \prod_v Z_v^{\deg(v)}$; Eulerianity proves $U_G^2 = \mathbb{I}$. For each party, measure the projectors onto complement pairs $\{|a_R\rangle, |\bar{a}_R\rangle\}$. These party-local projectors commute with U_G and resolve the Hilbert space into three-qubit blocks. The measurements cannot increase the average of Eq. (1): for each bipartition this follows from entropy concavity on the unmeasured side and purity, applied successively to all three local measurements.

On a block with encoded bits x, y, z,

$$q = q_0 + \ell_A x + \ell_B y + \ell_C z + e(xy + yz + zx) \pmod 2. \tag{5}$$

Since $q(x,y,z) + q(1-x,1-y,1-z) = |E|$ modulo two, $\sum_R \ell_R = e + |E|$ modulo two. When $e = 0$, Eq. (5) and the phase in Eq. (2) give a sign times a product of local X or Y operators. Their charge eigenspaces have product bases.

For $e = 1$, encoded local X operations change ℓ by any even-parity vector. If $|E|$ is odd first conjugate by an encoded $S = \mathrm{diag}(1, i)$, changing the linear parity; then reduce to $\ell = (1,0,0)$. Up to a sign and local unitary, every block is an eigenspace of

$$V = i X_A X_B X_C \,\mathrm{CZ}_{AB}\,\mathrm{CZ}_{BC}\,\mathrm{CZ}_{CA}, \quad V^2 = \mathbb{I}. \tag{6}$$

For an eigenstate, $V Z_R V^\dagger = -Z_R$ confines its marginal Bloch vector to the XY plane. Let its length be r_R, and

choose an XY Pauli observable M_R with expectation r_R. If $M_A = aX_A + bY_A$, then $VM_AV^\dagger = (aX_A - bY_A)Z_BZ_C$ anticommutes with M_B. Squaring a real linear combination gives

$$r_A^2 + r_B^2 \le 1, \quad r_B^2 + r_C^2 \le 1, \quad r_C^2 + r_A^2 \le 1. \tag{7}$$

The anticommuting-observable technique has prior uses in entropy uncertainty relations [5]. With $x_R = r_R^2$,

$$S(\rho_R) = f(x_R) = \log 2 - \sum_{j\ge 1} \frac{x_R^j}{2j(2j-1)}, \tag{8}$$

a concave function. The polytope in Eq. (7) has vertices 0, the three coordinate unit vectors, and $(1/2, 1/2, 1/2)$. Indeed, with $m = \min_R x_R$, its explicit convex decomposition is

$$\begin{aligned} x = (1 - \textstyle\sum_R x_R + m)0 + \sum_R (x_R - m)e_R \\ + 2m(1/2, 1/2, 1/2). \end{aligned} \tag{9}$$

The vertex values give B_w. The block measurements prove the bound for pure charge eigenstates, and every vector in an ensemble of a strong sector state lies in that sector, proving the mixed-state bound.

For attainment, let $U_{100}|xyz\rangle = (-1)^{x+xy+yz+zx}|\bar{x}\bar{y}\bar{z}\rangle$; it obeys $S_A V S_A^\dagger = -U_{100}$. In its $+1$ sector the normalized lexicographic vector

$$v_* = (1, 1, 1, i, -i, 1, 1, 1)/\sqrt{8} \tag{10}$$

has all marginal eigenvalues $p_*, 1 - p_*$. The four vectors obtained using $\mathbb{I}, Z_AZ_B, Z_AZ_C, Z_BZ_C$ are orthonormal. For the other branch, the following four vectors divided by 2 have C pure and A, B maximally mixed, and form a $+1$-sector basis:

$$\begin{aligned} &(1, 1, 0, 0, 0, 0, 1, 1), \quad (1, -1, 0, 0, 0, 0, -1, 1), \\ &(0, 0, 1, i, -i, 1, 0, 0), \quad (0, 0, 1, -i, i, 1, 0, 0). \end{aligned} \tag{11}$$

Permuting parties in Eq. (6) and applying a local Z supplies the other choices and the opposite sector. Assembling the appropriate bases over all blocks proves the equality in Eq. (4). These graph sectors are not generally Pauli stabilizer-code projectors; familiar graph-state cut-rank formulas [6] do not directly give this roof.

Finite-range anomaly.— For a chain take edges $(j, j + r)$ for r in a finite set S, with $R = \max S$, $L > 2R$, and each connected arc of length at least R. Then $e = \kappa := \sum_{r\in S} r \pmod 2$. To identify the endpoint class, restrict the unphased symmetry to an interval I much longer than R, keeping its internal edges. Its square is $(-1)^{|E(I)|}L_IR_I$, with

$$L_I = \prod_{j=1}^{R} Z_j^{\sum_{r\in S} \mathbf{1}_{j\le r} \bmod 2}, \qquad U_I L_I U_I^\dagger = (-1)^\kappa L_I. \tag{12}$$

The endpoint associator is therefore $(-1)^\kappa$ in the standard symmetry-localization construction [1, 4]. Thus $\kappa = 1$ realizes the nontrivial $H^3(\mathbb{Z}_2, U(1))$ class and has the sharp bound of Eq. (4). This is a statement about these representatives, not arbitrary finite-depth conjugates. An abstract Eulerian graph alone carries no assumed spatial anomaly.

$\mathbb{Z}_N$ boundary actions.— Let $N \ge 2$, $L \ge 3$ and $1 \le p < N$. On an oriented ring of N-level sites define

$$U|x\rangle = \exp\left[\frac{2\pi i p}{N^2} \sum_j [x_{j+1} - x_j]_N\right] |x + \mathbf{1}\rangle, \tag{13}$$

where $x = (x_1, \ldots, x_L)$, $\mathbf{1} = (1, \ldots, 1)$, and $[\cdot]_N \in \{0, \ldots, N-1\}$. The sum is divisible by N on the ring, so $U^N = \mathbb{I}$. This is a standard type-I boundary action [7, 8]; phase-branch conventions differ in the literature. The explicit polynomial branch of Ref. [8] is $U_{\rm pub} = JU^\dagger J^\dagger$, where the on-site unitary $J = \bigotimes_j (|x_j\rangle \mapsto |-x_j\rangle)$ maps the charge label q to $-q$. Hence all entanglement statements below also hold for that branch. For an interval $I = [\ell, r]$ the internal-bond restriction obeys $U_I^N = Z_r^p Z_\ell^{-p}$ and $U_I Z_r U_I^\dagger = \omega^{-1} Z_r$, with $\omega = e^{2\pi i/N}$. The endpoint cocycle is $\omega^{\pm pa\lfloor (b+c)/N \rfloor}$; its class is nonzero for $p \not\equiv 0 \pmod N$ [4, 8].

Every strongly symmetric state of Eq. (13) on three nonempty connected arcs satisfies the explicit size-independent bound

$$E_3(\rho) \ge \frac{\sin^4(\pi p/N)}{82944\, N^6}. \tag{14}$$

It is conservative, not an optimal constant. To see it, measure in each arc the orbits of strings under uniform $\mathbb{Z}_N$ shifts; the local projectors commute with U and leave three N-dimensional parties. For a product block vector ψ, set $z = \langle U\rangle_\psi$ and $\delta = \sqrt{2(1 - |z|)}$. A basis measurement of each marginal shows that its probabilities and their cyclic shift differ in ℓ_1 norm by at most 2δ. Thus each probability is at least $1/N - 2\delta$.

Relabel the two first arc shifts so their common bond has zero offset. The four phase factors g_{ab} at $a, b \in \{0, 1\}$ and fixed third shift satisfy $g_{00}g_{11}/(g_{10}g_{01}) = e^{-2\pi i p/N}$: only the common bond survives this mixed difference. If $\delta < \sin(\pi p/N)/[3(2N)^{3/2}]$, all local probabilities exceed $1/(2N)$. The four normalized component equations for an approximate eigenvector differ from the corresponding g_{ab} by at most $\eta = \delta(2N)^{3/2} < 1$. Since a product vector gives a factorized ratio, $2\sin(\pi p/N) \le 4\eta + 2\eta^2 \le 6\eta$, a contradiction. Consequently all fully tripartite-separable states obey

$$|\operatorname{Tr}(\rho U)| \le 1 - \frac{\sin^2(\pi p/N)}{144 N^3}. \tag{15}$$

For arbitrary ρ, the product of pure-state marginals is separable; quantum Pinsker and convexity of the

squared positive part give $E_3(\rho) \geq \frac{1}{4}[|\operatorname{Tr}(\rho U)| - 1 + \sin^2(\pi p/N)/(144N^3)]_+^2$. In a charge sector $|\operatorname{Tr}(\rho U)| = 1$, yielding Eq. (14). The orbit measurements are local with respect to the selected parties; they need not be spatially local within an arc.

NPT sector projectors.— For the same action let

$$\rho_q = \frac{1}{N^L} \sum_{k=0}^{N-1} \omega^{-qk} U^k, \qquad q = 0, \ldots, N-1. \tag{16}$$

Its rank is N^{L-1}. Across any bipartition into two nonempty connected arcs, local uniform-shift orbits give $N \times N$ blocks. Choose an orbit pair with zero endpoint holonomy; its block action is locally equivalent to $X \otimes X$ times a diagonal phase $\gamma(d)$ depending on the relative shift d, where $\gamma(0) = 1$ and $\gamma(d \neq 0) = t := \omega^p$. Local clock rotations change the charge, so take $q = 0$. In the partial transpose, the subspace of fixed $a + b = 1$ modulo N is invariant. For $N \geq 3$, its principal minor on indices $a = 0, 1, 2$ is

$$\frac{1}{N^2} \begin{pmatrix} 1 & 1 & t^2 \\ 1 & 1 & t \\ t^{-2} & t^{-1} & 1 \end{pmatrix}, \quad \det = -\frac{4\sin^2(\pi p/N)}{N^6} < 0. \tag{17}$$

This orbit block occurs for every connected cut. Thus every ρ_q is NPT for $N \geq 3$ and $p \neq 0$, including noncoprime p, N. Moreover, the zero-holonomy blocks carry total weight $1/N$ for $L \geq 3$. Writing $\mathcal{N}(\rho) = (\|\rho^{T_A}\|_1 - 1)/2$ and $d = t^2 - t$, the minor's Rayleigh vector $(1, -1, -\bar{d})$ gives the size-independent bound

$$\mathcal{N}(\rho_q) \geq \frac{4\sin^2(\pi p/N)}{N^3[2 + 4\sin^2(\pi p/N)]} > 0. \tag{18}$$

An NPT sector projector cannot be a mixture of bipartite-product eigenvectors, so no complete orthonormal product eigenbasis exists. In fact even a nonorthogonal product eigenbasis is impossible: any product eigenvector in a zero-holonomy block must have full cyclic support and would force the relative-phase matrix $D_{ab} = 1$ for $a = b$ and t otherwise to have rank one. Its rows $(0, 1)$ and columns $(0, 2)$ instead give the nonzero minor $t - t^2$. Local orbit projections commute with U and preserve product form, so all global product eigenvectors have zero projection onto these blocks and cannot span the full space.

In fact, its best separable weight across this cut [11] is exactly $1/N$. To see this, the N equally weighted holonomy classes have, up to a scalar, $\gamma_h(d) = t^{\mathbf{1}_{d>h}}$ for $h = 0, \ldots, N-1$. A product eigenvector requires γ_h to be a character of $\mathbb{Z}_N$. For $h = 0$, $\gamma_h(1) = \gamma_h(2) = t$ contradicts the character law; for $1 \leq h \leq N-2$, $\gamma_h(1) = 1$ but $\gamma_h(h+1) = t \neq 1$. Only $h = N-1$ admits a product eigenbasis. Any separable component in a positive decomposition of ρ_q is supported on its charge sector, and hence on these good blocks, which carry weight $1/N$. Their explicitly separable mixture attains that weight.

This contradicts the broad finite-Abelian extension expected after Eq. (23) of Ref. [1]. That expectation is not a theorem; no claim is made about other symmetry representatives or about every strong-sector state being bipartite entangled. Each proper subsystem of ρ_q is exactly maximally mixed, because the partial trace of U^k vanishes for $1 \leq k < N$.

For $N = 3$, $p = 1, 2$, the connected-cut orbit blocks have three equally frequent holonomies. Up to phases their relative-shift diagonals are $(1, t, t)$, $(1, 1, t)$, and $(1, 1, 1)$, with $t = \omega^p$; the $p = 2$ phases are complex conjugates of $p = 1$. The last gives a separable block. Each of the other two partial transposes splits into three identical-spectrum 3×3 blocks; after multiplication by 9 their diagonals and off-diagonal magnitudes are one, and their triangular phase products are $\omega^{\pm 1}$. Their characteristic polynomial is $x^3 - 3x^2 + 3$. Thus

$$\|\rho_q^{T_A}\|_1 = \frac{5 + 8\cos(\pi/9)}{9}, \qquad \mathcal{N}(\rho_q) = \frac{4\cos(\pi/9) - 2}{9}. \tag{19}$$

The qutrit orbit blocks are equal mixtures of three Bell states. Their line-versus-triangle separability pattern and the relevant partial transpose analysis have close prior work [9, 10]. In particular the cubic above follows by specializing the general two-qutrit partial-transpose formula of Ref. [9]; our calculation identifies these blocks and their weights in the many-site sector.

Scope and status.— The numerical bounds concern the displayed representatives and partitions, not arbitrary representatives of the same anomaly class. The graph calculation alone does not assume spatial locality; Eq. (12) uses the stated finite-range geometry. Ref. [1] already proves qualitative anomaly-imposed tripartite nonseparability and separately establishes a preparation obstruction. Our block measurements do not establish a new finite-depth result. Numerical and exact finite-size checks support, but do not replace, the analytic arguments. Specialist review and a complete priority assessment remain open. OpenAI Codex (including GPT-6.1 Terra subagents) assisted derivation, literature search, analytic and computational checks, and drafting under user direction; independent human verification and author identification are pending.

[1] L. A. Lessa, M. Cheng, and C. Wang, Mixed-state quantum anomaly and multipartite entanglement, Phys. Rev. X **15**, 011069 (2025), arXiv:2401.17357.

[2] X. Chen, Z.-X. Liu, and X.-G. Wen, 2D symmetry protected topological orders and their protected gapless edge excitations, Phys. Rev. B **84**, 235141 (2011), arXiv:1106.4752.

[3] Sz. Szalay, Multipartite entanglement measures, Phys. Rev. A **92**, 042329 (2015), arXiv:1503.06071.

[4] D. V. Else and C. Nayak, Classifying symmetry-protected topological phases through the anomalous action of the symmetry on the edge, Phys. Rev. B **90**, 235137 (2014), arXiv:1409.5436.

[5] S. Wehner and A. Winter, Higher entropic uncertainty relations for anti-commuting observables, J. Math. Phys. **49**, 062105 (2008), arXiv:0710.1185.

[6] M. Hein, J. Eisert, and H. J. Briegel, Multiparty entanglement in graph states, Phys. Rev. A **69**, 062311 (2004), arXiv:quant-ph/0307130.

[7] L. H. Santos and J. Wang, Symmetry-protected many-body Aharonov-Bohm effect, Phys. Rev. B **89**, 195122 (2014), arXiv:1310.8291.

[8] J. C. Wang, L. H. Santos, and X.-G. Wen, Bosonic anomalies, induced fractional quantum numbers and degenerate zero modes, Phys. Rev. B **91**, 195134 (2015), arXiv:1403.5256.

[9] B. Baumgartner, B. C. Hiesmayr, and H. Narnhofer, The state space for two qutrits has a phase space structure in its core, Phys. Rev. A **74**, 032327 (2006), arXiv:quant-ph/0606083.

[10] L. Derkacz and L. Jakóbczyk, Entanglement of a class of mixed two-qutrit states, Phys. Rev. A **76**, 042304 (2007), arXiv:0707.1575.

[11] M. Lewenstein and A. Sanpera, Separability and entanglement of composite quantum systems, Phys. Rev. Lett. **80**, 2261 (1998), arXiv:quant-ph/9707043.